# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

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**COURSE NAME: INTERNET PROGRAMMING** **Marks: 81**

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**COURSE OUTCOME COVERED**

**CO-1:** Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery.**(L2)**

Assessment Tool:Code along exercise: Make your first HTML page together with CSS

**Weightage: 10%**

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**QUESTIONS:** **Total Marks: 10**

## Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

## Data Snippets Provided

**Snippet 1: Video Embedding Options** A.
```
<media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

**Snippet 2: Form Submission Methods**

```
A. GET
B. POST
C. PUT
D. DELETE
```

**Snippet 3: Navigation Elements**

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

**Snippet 4: XML Syntax**

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

**Snippet 5: Web Communication Protocols**

```
A. FTP
B. HTTP  C. SMTP
D. SNMP
```

## Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

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***Code of Conduct:***

*I* Navuluri Anjali (192324137) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:* **NAVULURI ANJALI**

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning | Marks |
|---|---|---|---|---|---|---|
| Understanding of Case | 25 | Thorough understanding; clearly identifies all key issues | Good understanding with most issues identified | Basic understanding ; some issues missed | Poor understanding; problem not identified correctly | 21 |
| Application of Concepts | 25 | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps | Limited application of concepts | Fails to apply relevant concepts | 21 |
| Analysis | 20 | In-depth analysis with strong reasoning and insights | Good analysis with logical reasoning | Basic analysis with limited depth | Weak or no analysis; lacks reasoning | 16 |
| Solution Design | 20 | Innovative, feasible solutions with strong justification | Logical solutions with adequate justification | Basic solutions with weak justification | Incorrect or no solution provided | 15 |
| Clarity | 10 | Clear, well-structured, and easy to understand | Organized and understandable presentation | Limited clarity and structure | Poor clarity; difficult to understand | 8 |
| | | | | | | 81/100 |

## ANSWERS AND JUSTIFICATIONS

**Question 1: Select the correct HTML tag for video embedding and justify your answer.**

**Answer:** Option B — <video src="video.mp4" controls></video>

**Justification:** The HTML <video> element is used to embed video content in a webpage. The src attribute specifies the video file, and the controls attribute displays controls such as play, pause, and volume. The other options are not standard HTML elements for embedding video.

**Detailed Explanation:** Before HTML5, videos had to be embedded through third-party plug-ins such as Adobe Flash using the <object> or <embed> tags, which required users to install extra software and caused security and compatibility problems. HTML5 introduced the <video> element so that browsers can play video natively. For the start-up's promotional video on the homepage, this means the video will work on desktops, phones, and tablets without any plug-in, and it can be styled with CSS and controlled with JavaScript.

The src attribute gives the location of the video file (here, video.mp4). The controls attribute is a Boolean attribute: when it is present, the browser shows its default playback interface with play/pause, a progress bar, volume, and full-screen options. Without controls, the visitor would see a video with no way to start or pause it unless the developer wrote custom controls. Other useful attributes include width and height (to reserve space on the page), poster (a preview image shown before playback), autoplay, loop, and muted. Fallback text can also be placed between the opening and closing tags so that very old browsers display a message instead of a blank area.

**Why the other options are incorrect:**

**Option A (<media>):** There is no <media> element in the HTML specification, so browsers ignore it and the video will not be displayed.

**Option C (<movie>):** <movie> is not a valid HTML element; it is not recognised by any standard browser.

**Option D (<vid>):** <vid> is only an invented abbreviation and is not part of HTML, so it has no effect.

**Conclusion:** Option B is the only standards-compliant choice. It follows the HTML5 specification, gives built-in playback controls, and works across modern browsers, which fits the requirement that the homepage video be embedded correctly.

**Example:**

```
<video src="video.mp4" controls width="640" height="360" poster="preview.jpg">
Your browser does not support the video tag. </video>
```

## Question 2: Identify the appropriate HTTP method for form submission and explain why.

**Answer:** Option B — POST

**Justification:** The POST method is appropriate for submitting registration form data to a server because the data is sent in the request body rather than being appended to the URL as it typically is with GET. PUT is commonly used to create or replace a resource, while DELETE is used to request deletion of a resource. POST alone does not encrypt data, so HTTPS should be used to protect sensitive information during transmission.

**Detailed Explanation:** HTTP defines several request methods, each with a specific purpose. A registration form collects personal details such as a username, password, and e-mail address, and then asks the server to create a new user account. This is an action that changes data on the server, so POST is the correct method. With POST, the form fields travel in the body of the HTTP request, which means they are not displayed in the browser address bar, are not stored in browser history, and are not normally written into server access logs as part of the URL.

With GET, the form data is appended to the URL as a query string (for example,

/register?username=anjali&password=1234). The URL can then be bookmarked, shared, cached, or recorded in logs, which exposes sensitive data. GET also has practical limits on URL length, and it is meant for retrieving data, not for changing it. POST has no such length restriction and can carry larger data, including file uploads. Because POST requests are not idempotent, browsers also warn the user before re-submitting a POST form, which helps prevent accidental duplicate registrations.

**Why the other options are not suitable:**

**Option A (GET):** It exposes form data in the URL and is intended for reading data, not for submitting sensitive information.

**Option C (PUT):** It is used to create or completely replace a specific resource at a known URL, and standard HTML forms do not support it directly.

**Option D (DELETE):** It is used to remove a resource and has no role in submitting registration data; HTML forms also do not support it directly.

**Security note:** POST by itself does not encrypt data. The request body can still be read in transit over an unsecured connection, so the registration page must be served over HTTPS (HTTP secured with TLS). The server should also validate and sanitise all input and store passwords only as salted hashes. This combination satisfies the requirement that the form must securely send user data to the server.

**Example:**

```
<form action="/register" method="POST">
<input type="text" name="username"
placeholder="Username" required>    <input
type="password" name="password"
placeholder="Password" required>    <button
type="submit">Register</button> </form>
```

## Question 3: Choose the proper semantic element for navigation and justify its usage.

**Answer:** Option B — <nav>

**Justification:** The HTML <nav> element represents a section containing major navigation links. It provides semantic meaning to the webpage structure and helps browsers, assistive technologies, and developers identify the navigation section. A  is a generic container, a <section> groups related content, and a <header> represents introductory content or a group of navigational aids.

**Detailed Explanation:** Semantic HTML means choosing elements according to the meaning of the content they hold, rather than only for how they look. HTML5 added elements such as <header>, <nav>, <main>, <section>, <article>, <aside>, and <footer> to describe the structure of a page. The <nav> element is designed for a block of major navigation links, such as the main menu, a table of contents, or pagination links. Since the case study requires the navigation menu to follow semantic HTML standards, <nav> is the correct choice.

Using <nav> brings several benefits. First, screen readers expose it as a navigation landmark, so visually impaired users can jump directly to the menu or skip it, which improves accessibility. Second, search engines can better understand the page structure, which supports SEO. Third, the code becomes easier to read, maintain, and style, because a developer immediately knows what the block is for. It is normally written with an unordered list of links inside, which is the most common and accessible pattern. Not every group of links needs <nav>; it is meant for the primary navigation blocks of the page.

**Why the other options are not the best choice:**

**Option A ( ):** It is a generic, non-semantic container. It gives no information about its content, so assistive technologies cannot recognise it as navigation.

**Option C (<section>):** It groups thematically related content, usually with a heading, and does not specifically mean navigation.

**Option D (<header>):** It represents introductory content for a page or section. It may contain a <nav> element, but it is not itself the navigation element.

**Conclusion:** <nav> is the standards-compliant, accessible, and meaningful element for a navigation menu, and it is therefore the correct answer.

**Example:**

```
<header>
    <nav aria-label="Main navigation">
        <ul>
            <li><a href="index.html">Home</a></li>
            <li><a href="about.html">About</a></li>
            <li><a href="contact.html">Contact</a></li>
        </ul>
    </nav>
</header>
```

### Question 4: Identify the correct XML syntax and explain the error in the other options.

**Answer:** Option B — <note><to>User</to><from>Admin</from></note>

**Justification:** Option B is correct because all XML elements have matching opening and closing tags, the elements are properly nested, and the document contains a single root element.

**Detailed Explanation:** XML (eXtensible Markup Language) is used to store and transport data, including configuration files. Unlike HTML, XML is strict: a document must be well-formed or a parser will reject it. The main rules are: (1) every opening tag must have a matching closing tag; (2) elements must be properly nested, so the element opened last must be closed first; (3) there must be exactly one root element that contains all the others; (4) tag names are case-sensitive; and (5) attribute values must be enclosed in quotes. Option B satisfies all of these rules. Here <note> is the root element, and it contains the child elements <to> and <from>, each correctly opened and closed.

**Explanation of the other options:**

**Option A:** The root <note> element is not closed. The structure <note><to>User</to><from>Admin</from> ends without a </note> tag, so the document is not wellformed and the parser will report an error such as "premature end of file" or "element not closed".

**Option C:** The <to> element is incorrectly closed using </from>, resulting in mismatched tags. The element opened with <to> must be closed with </to>. In addition, there is no <from> opening tag at all, so the closing </from> has nothing to match.

**Option D:** The opening <note> tag is incomplete because the closing angle bracket (>) is missing. The text <note to="User"></note ends without a closing > on the end tag, so the tag is not complete and the parser cannot read it. (The attribute to="User" is itself written correctly with quotes.)

**Conclusion:** Only Option B is a well-formed XML document. Since the team's XML configuration files contain syntax errors, checking these basic rules (matching tags, correct nesting, a single root, and complete tags) is the first step in fixing them.

**Correct XML example:**

```
<?xml version="1.0" encoding="UTF-8"?>
<note>
    <to>User</to>
```

```
    <from>Admin</from>
</note>
```

### Question 5: Choose the correct protocol for web communication and justify your choice.

**Answer:** Option B — HTTP

**Justification:** HTTP stands for Hypertext Transfer Protocol. It is an application-layer protocol used for communication between web clients, such as browsers, and web servers. It enables browsers to request webpages and servers to deliver resources such as HTML documents, images, and other content.

**Detailed Explanation:** HTTP works on a request–response model. When a user types a web address or clicks a link, the browser (the client) sends an HTTP request to the web server. The request contains a method such as GET or POST, the path of the resource, and headers. The server processes it and sends back an HTTP response containing a status code (for example, 200 OK, 404 Not Found, or 500 Internal Server Error), response headers, and the requested content, such as an HTML page, CSS file, image, or JSON data. HTTP is stateless, meaning each request is independent, so web applications use cookies or sessions to remember users between requests. It normally runs over TCP on port 80, while its secure version, HTTPS, uses port 443.

Since the start-up is building a modern web application that must deliver web content and receive form submissions (like the registration form in this case), HTTP is the protocol that connects the browser to the server. All the other topics in this case study, namely HTML pages, video delivery, form submission with POST, and XML data exchange, are carried over HTTP.

**Explanation of the other options:**

**FTP:** File Transfer Protocol is used for uploading and downloading files between systems. It is not used by browsers to request and display webpages.

**SMTP:** Simple Mail Transfer Protocol is used for sending e-mail between mail servers and from mail clients. It has no role in loading webpages.

**SNMP:** Simple Network Management Protocol is used for monitoring and managing network devices such as routers, switches, and printers. It is not used for web content delivery.

**Note:** HTTPS is HTTP secured using TLS and is generally preferred for modern web applications because it protects data transmitted between the browser and server. It encrypts the communication, verifies the identity of the server through certificates, and protects against tampering, which is especially important for login and registration forms.